

Papers

Wednesday, July 15 | 11:30–13:00

RWA BP

Corcovado

Chair: Alexander Brownlee, *University of Stirling, United Kingdom*

- 11:30–11:50 **Robust Multi-Objective Optimization for Bicycle Rebalancing in Shared Mobility Systems** 
Diego Daniel Pedroza-Perez, Gabriel Luque, Sergio Nesmachnow, Jamal Toutouh  
- 11:50–12:10 **Evolving Macro-Actions for Monte Carlo Tree Search in Real-Time Domains** 
Andrew Gourley, Michael Dann, Xiaodong Li, Fabio Zambetta  
- 12:10–12:30 **★ Evolution With Purpose: Hierarchy-Informed Optimization of Whole-Brain Models** 
Hormoz Shahrzad, Niharika Gajawelli, Kaitlin Maile, Manish Saggar, Risto Miikkulainen  
- 12:30–12:50 **★ From Prompts to Pareto Fronts: Using Code LMs as Semantic Genetic Operators for Compiler Bug Discovery** 
Lang Nguyet Anh, Ho Duc Luong, Vu An  

GA + Theory BP

Manuel Antonio

Chair: Carlos A. Coello Coello, *CINVESTAV-IPN, Department of Computer Science, Mexico*

- 11:30–11:50 **Analysis of the Impact of Tournament Size on Coevolutionary Search** 
Alistair Benford, Per Kristian Lehre  
- 11:50–12:10 **Gray-Box Optimization and the Vertex Coloring Problem** 
Johanna Gasse, Antonia Heinen, Hendrik Higl, Timo Kötzing  
- 12:10–12:30 **★ Dropout-Inspired Strategies For Enhanced Exploration In Evolutionary Algorithms** 
Daniele Ganci, Luís Correia  
- 12:30–12:50 **★ Parent Selection Mechanisms in Elitist Crossover-Based Algorithms** 
Andre Opris, Denis Antipov  

CS + SI BP

Poas

Chair: Peter A.N. Bosman, *Centre for Mathematics and Computer Science, Delft University of Technology, Netherlands*

- 11:30–11:50 **Dynamic Descriptor Mutations Improve Automated Quality-Diversity** 
Reece Van Der Bank, Geoff Nitschke  
- 11:50–12:10 **★ Automated PSO Configuration Framework via Deep Neural Networks and Function Landscape Descriptors** 
Yutong Zou, Peter Bentley  
- 12:10–12:30 **★ In Search of the Ingredients of Open-Endedness: Replicating Picbreeder with Large Vision-Language Models** 
Sam Earle, Kai Arulkumaran, Akarsh Kumar, Andrew Dai, Julian Togelius, Sebastian Risi  
- 12:30–12:50 **Stability-Driven CPG Design for More Effective Quality-Diversity in Evolutionary Robotics** 
Matthias Pex, Dries Marzougui, Francis wyffels  

BBSR 1

Monteverde

Chair: Mike Preuss, *Universiteit Leiden, Netherlands*

- 11:30–11:50 **Benchmarking Evolutionary and Machine Learning Algorithms: The SIREN Challenge of Emergency Medical Services** 
Torjus Kallekleiv, Xavier Sánchez-Díaz, Ole Mengshoel  
- 11:50–12:10 **Block-Bench: A Framework for Controllable and Transparent Discrete Optimization Benchmarking** 
Furong Ye, Frank Neumann, Thomas Bäck, Niki van Stein  
- 12:10–12:30 **Benchmarking Stopping Criteria for Evolutionary Multi-objective Optimization** 
Kenji Kitamura, Ryoji Tanabe  

EMO 1

Tortuguero

Chair: Deepanshu Yadav, *Michigan State University, United States*

- 11:30–11:50 **Dominance-Boundary-Based Reference Point Generation for Indicator-Based Multiobjective Optimization** 
Kentaro Matsumoto, Hiroyuki Sato  
- 11:50–12:10 **Not All Problems Are Equal: Weighted Performance Profiles For Many-Objective Optimization** 
Regina Carla Sousa, Denis E. Vargas, Elizabeth Wanner, Joshua Knowles  
- 12:10–12:30 **Stable Normalization for Preference-based Subset Selection with an Unbounded External Archive** 
Yuta Nakanishi, Mamoru Doi, Yusuke Yasuda, Hiroyuki Sato  
- 12:30–12:50 **Simultaneous Approximation of Constraint-Specific Pareto Fronts by Constraint-Decomposed Push and Pull Search** 
Ryo Takamiya, Minami Miyakawa, Keiki Takadama, Hiroyuki Sato  

ECOM 1

Chirripó

Chair: Francisco Chicano, *University of Málaga, Spain*

- 11:30–11:50 **A Granular Ball-Ant Colony Optimization Framework Enhanced by Variable Neighborhood Search for the Multi-Depot Half-Open Time-Dependent EVRP** 
Yingkai Xu, Anikó Kopacz, Camelia Chira  
- 11:50–12:10 **Adaptive Stochastic Natural Gradient Method for Safe Optimization on Binary Space** 
Kento Uchida, Ryoki Hamano, Masahiro Nomura, Shinichi Shirakawa  
- 12:10–12:30 **Minimizing restacks in load-unload-ordered rectangular packing via Generational Sequence Pair** 
Hidenori Ohta  
- 12:30–12:50 **Addressing Combinatorial Optimization with Estimation of Distribution Algorithms Based on Diffusion Models** 
Roberto Santana, Jose Lozano  

EML 1

Gandoca-Manzanillo

Chair: Will Neil Browne, *Queensland University of Technology, Victoria University of Wellington, Australia; New Zealand*

- 11:30–11:50 **Investigating Robustness in Vision-Language Models via Adversarial Prompt Illumination** 
Thai Huy Nguyen, Khoa Tran, Quan Minh Phan, Ngoc Hoang Luong  
- 11:50–12:10 **Wave-GA: Cyclical Evolutionary Optimization for Streaming Classification with Emerging Labels** 
Giannis Vassiliou, Georgia Trouli, Sotirios Batsakis, Haridimos Kondylakis  

- 12:10–12:30 **Phase-Alternation and Tree-Initialization to Facilitate Interpretable Rule-based Machine Learning** 
Gabriel Lipschutz-Villa, Harsh Bandhey, Khoi Dinh, Michael Heider, Malek Kamoun, Ryan Urbanowicz  
- 12:30–12:50 **Multi-Objective Evolution of Patch Tokens for Tabular Transformers** 
Han Joon Byun, Joon Huh, Jieun Yook, Byung-ro Moon  

Wednesday, July 15 | 15:00–16:30

RWA 1

Corcovado

Chair: Daniel R. Tauritz, *Auburn University, United States*

- 15:00–15:20 **Should I Overtake? Cue Learning using Evolution for Accurate Recognition of Safe Autonomous Vehicle Maneuvers** 
Peter Bentley, Soo Ling Lim, Fuyuki Ishikawa, Paolo Arcaini  
- 15:20–15:40 **BACE: LLM-based Code Generation through Bayesian Anchored Co-Evolution of Code and Test Populations** 
Kaushitha Silva, Srinath Perera  
- 15:40–16:00 **Many-Objective Multiple TSP in Path-Influenced Environments using Evolutionary Optimization** 
Carlo Nübel, Sanaz Mostaghim  
- 16:00–16:20 **Landscape-aware Automated Algorithm Design: An Efficient Framework for Real-world Optimization** 
Haoran Yin, Shuaiqun Pan, Zhao Wei, Jian Cheng Wong, Yew-Soon Ong, Anna Kononova, Thomas Bäck, Niki van Stein  

Theory 1

Manuel Antonio

Chair: Michal Witold Przewozniczek, *Wroclaw University of Science and Technology, Poland*

- 15:00–15:20 **Anytime Analysis on BinVal: Adaptive Parameters Help** 
Timo Kötzing, Jurek Sander  
- 15:20–15:40 **Convergence Analysis of Evolution Strategies for Mixed-Integer Optimization** 
Ryoki Hamano, Kento Uchida, Shinichi Shirakawa  
- 15:40–16:00 **When Switching Algorithms Helps: A Theoretical Study of Online Algorithm Selection** 
Denis Antipov, Carola Doerr  
- 16:00–16:20 **A Generic Framework for Optimisation under Uncertainty with Recourse: Theory and Examples** 
Joshua Knowles, Per Kristian Lehre, Shishen Lin, Frank Neumann, Janina Schreiber, Christine Zarges  

GP 1

Tortuguero

Chair: Jose Manuel Velasco, *Universidad Complutense, Bioinspired Intelligence S.L., Spain*

- 15:00–15:20 **From Six to Two: SLIM-DUO, a Simplified Extension of SLIM-GSGP** 
Liah Rosenfeld, Leonardo Vanneschi  
- 15:20–15:40 **Prior-Informed Symbolic Regression for Multiview Scientific Discovery** 
Hernan Lira, Luis Martí, Nayat Sanchez-Pi  
- 15:40–16:00 **Fitness Landscape Analysis of Grammatical Evolution Models for Glucose Prediction** 
J. Manuel Velasco, Daniel Parra, Óscar Garnica, J. Ignacio Hidalgo  

ECOM 2

Chirripó

Chair: Luca Mariot, *Semantics, Cybersecurity & Services Group, University of Twente, The Netherlands, Netherlands*

- 15:00–15:20 **When Frequency Fitness Assignment Fails: Trapped States in Frequency-Guided Local Search** 
Jiazheng Zeng, Thomas Weise, Zhize Wu, Markus Wagner  
- 15:20–15:40 **Digging to the Ground Truth: Solving Multi-objective Gray-Box Optimization Problems through Hyperplane Elimination** 
Carolyn Mensendiek, Oliver Preuß, Jeroen Rook, Francisco Chicano, Darrell Whitley, Heike Trautmann  
- 15:40–16:00 **Accelerating LLM-Based Algorithm Evolution for the 3D Container Loading Problem** 
Guorui Quan, Mingfei Sun, Manuel López-Ibáñez, Nicolás Álvarez-Gil, Silvino Fernández Alzueta  
- 16:00–16:20 **GRASP-RVND Metaheuristic for the Traveling Salesman Problem with Drone Launch Station Location** 
Tarcisio Barroso Marques, Jose Elias Claudio Arroyo  

EML 2

Gandoca-Manzanillo

Chair: Ryan Urbanowicz, *Cedars-Sinai Medical Center, United States*

- 15:00–15:20 **Cooperative Coevolution versus Monolithic Evolutionary Search for Semi-Supervised Tabular Classification** 
Jamal Toutouh  
- 15:20–15:40 **Multi-Objective Orchestration of Small Language Model Ensembles: Balancing Accuracy, Diversity, and Fairness** 
Advay Dhar, Swagatam Das, Kalyanmoy Deb  
- 15:40–16:00 **Personalized Federated Multi-Task Optimization via Learning-assisted Search Path Reconstruction** 
Zhenhai Yang, Binhong Xie, Lihu Pan, Jie Tian, Hui Liu, Xiaoxu Zhang, Chen Wu  
- 16:00–16:20 **SONATA: Self-adaptive Evolutionary Framework for Hardware-aware Neural Architecture Search** 
Halima Bouzidi, Smail Niar, Hamza Ouarnoughi, El-Ghazali Talbi  

Thursday, July 16 | 11:30–13:00**GP BP**

Corcovado

Chair: Nelishia Pillay, *University of Pretoria, South Africa*

- 11:30–11:50 **A Comparative Study of Model Selection Criteria for Symbolic Regression** 
Ali Soltani, Gabriel Kronberger, Fabricio Olivetti de Franca, Mattia Billa, Alessandro Lucantonio  
- 11:50–12:10 **Evolving Hardware-Efficient Grover Circuits with Grammatical Evolution** 
Arinze Obidiegwu, Douglas Mota Dias, Emmanuel Obidiegwu, Conor Ryan  
- 12:10–12:30 **★ Introns and Templates Matter: Rethinking Linkage in GP-GOMEA** 
Johannes Koch, Tanja Alderliesten, Peter Bosman  
- 12:30–12:50 **★ Generalization Analysis of Symbolic Regression Algorithms via Stability Theory** 
Kei Sen Fong, Mehul Motani  

EMO BP

Manuel Antonio

Chair: Carlos Ignacio Hernández Castellanos, *IIMAS-UNAM, Mexico*

- 11:30–11:50 **A Bi-Criteria Selection Framework for Incorporating Preference Information into Evolutionary Multi-objective Optimization** 
Ryuichi Mogami, Ryoji Tanabe  
- 11:50–12:10 **★ Structural Bias in Continuous Multi-objective Optimization** 
Jakub Kudela, Niki van Stein, Thomas Bäck, Anna Kononova  
- 12:10–12:30 **★ Multi-objective Local Optima Networks based on Decomposition** 
Gabriela Ochoa, Quentin Renau, Arnaud Liefooghe, Jonathan Fieldsend  
- 12:30–12:50 **★ QUIVER: Cost-Aware Adaptive Preference Querying in Surrogate-Assisted Evolutionary Multi-Objective Optimization** 
Florian Burnat  

NE BP

Poas

Chair: Penousal Machado, *University of Coimbra, CISUC, Portugal*

- 11:30–11:50 **★ Evolving Multi-Channel Confidence-Aware Activation Functions for Missing Data with Channel Propagation** 
Naeem Shahabi Sani, Ferial Najiantabriz, Shayan Shafaei, Dean Hougen  
- 11:50–12:10 **Optimizing Chlorination in Water Distribution Systems via Surrogate-assisted Neuroevolution** 
Rivaaj Monsia, Daniel Young, Olivier Francon, Risto Miikkulainen  
- 12:10–12:30 **LsmGrid: A Data-Aware Neuroevolution Framework for Designing Minimal Liquid State Machines** 
Andrés Romo, Andrés Otero, José M. Lanza-Gutierrez  
- 12:30–12:50 **NEAT-DNFs: A NeuroEvolutionary Framework for Evolving Dynamic Neural Field Architectures** 
João Cunha, Wolfram Erlhagen, Raymond Cuijpers, Estela Bicho  

ENUM 1

Arenal

Chair: Pierrick Legrand, *Bordeaux INP, ENSC, F-33400 Talence, France; nria Center of the University of Bordeaux, France, France*

- 11:30–11:50 **Effect of Mirrored Orthogonal Sampling for CMA-ES on Multimodal Problems** 
Kaito Nakatani, Masahiro Nomura, Isao Ono  
- 11:50–12:10 **Hierarchical Evolution Strategy for Optimization of Sharp Ridge** 
Sota Hamada, Yutaro Yamada, Kento Uchida, Shinichi Shirakawa  
- 12:10–12:30 **Evaluation of Element-wise Effectiveness Estimation for Augmented Lagrangian CMA-ES** 
Haruhito Nakagawa, Kento Uchida, Shinichi Shirakawa  
- 12:30–12:50 **On the Use of the "Mutate Large, But Inherit Small Principle" in Global and Noisy Optimization** 
Lisa Schönnenberger, Hans-Georg Beyer  

GA 1

Monteverde

Chair: Dirk Thierens, *Utrecht University, Netherlands*

- 11:30–11:50 **Evolving Robust Schedules: Advantages of Genetic Algorithms for FJSSP-W under Uncertainties** 
David Hutter, Thomas Steinberger, Michael Hellwig  
- 11:50–12:10 **Limited Perfect Monotonical Surrogates constructed using low-cost recursive empirical linkage discovery with guaranteed output** 
Michał Przewozniczek, Francisco Chicano, Marcin Komarnicki, Renato Tinós  

- 12:10–12:30 **Evolutionary Algorithms for Generating Graphs Matching Desired Laplacian Spectra** 
Hendrik Richter, Frank Neumann  

L4EC 1

Tortuguero

Chair: Eva Tuba, *Trinity University, United States*

- 11:30–11:50 **On the Sufficiency of SAT Features for Genetic Algorithm Parameter Selection** 
Paweł Kolendo, Wojciech Chmiel, Michal Pluháček  
- 11:50–12:10 **Similarity-based Portfolio Construction for Black-box Optimization** 
Catalin-Viorel Dinu, Diederick Vermetten, Carola Doerr  
- 12:10–12:30 **Analyzing search behavior of population-based optimization algorithms using graph neural networks** 
Bartłomiej Walczak, Hubert Guzowski, Wojciech Achtelek, Maciej Smolka, Jacek Mańdziuk  
- 12:30–12:50 **MODE: Manifold Operator Decoding Embeddings for Neural Neighborhood Search for Vehicle Routing Problems** 
Xiangling Chen, Yi Mei, Mengjie Zhang  

RWA 2

Chirripó

Chair: Alexander Brownlee, *University of Stirling, United Kingdom*

- 11:30–11:50 **Exact and Heuristic Methods for Capacitated Path Planning for Agricultural Field Operations** 
Fabliha Zahin, Shahadat Hossain  
- 11:50–12:10 **To Balance Competitive Games Using a Multi-objective Coevolutionary Approach** 
Shashank Raj, Auden Garrard, Ryan McKendrick, Bradley Feest, Kalyanmoy Deb  
- 12:10–12:30 **Operational Decision Support for Evacuating Nursing Home Residents: A Hamburg Benchmark and Metaheuristic Comparison Under a 5-Minute Time Budget** 
Nick Alexander, Milva Tannenbaum, Jörg Noennig  
- 12:30–12:50 **IDEM Enough? Evolving Highly Nonlinear Idempotent Boolean Functions** 
Claude Carlet, Marko Djurasevic, Domagoj Jakobovic, Luca Mariot, Stjepan Picek  

EML 3

Gandoca-Manzanillo

Chair: Will Neil Browne, *Queensland University of Technology, Victoria University of Wellington, Australia; New Zealand*

- 11:30–11:50 **Evolutionary Mapping of Neural Networks to Spatial Accelerators** 
Alessandro Pierro, Jason Yik, Jonathan Timcheck, Marius Lindauer, Eyke Hüllermeier, Marcel Wever  
- 11:50–12:10 **Surrogate-Assisted Evolutionary Multi-Agent Reinforcement Learning with Adaptive Fitness Evaluation** 
Cong Yu, Yuxing Wang, Xinyang Li, Zaihui Yang, Haoyu Wang, JunBo Tan, Yongzhe Chang, Tiantian Zhang, Xueqian Wang  
- 12:10–12:30 **Value Consistency Prioritization for Accelerating Knowledge Discovery in Sparse Reward Anticipatory Classifier Systems** 
Olgierd Unold, Jan Zemło  
- 12:30–12:50 **Evolutionary Discovery of Reinforcement Learning Algorithms via Large Language Models** 
Alkis Sygkounas, Amy Loutfi, Andreas Persson  

Thursday, July 16 | 15:00–16:30

EML BP

Corcovado

Chair: Ryan Urbanowicz, *Cedars-Sinai Medical Center, United States*

- 15:00–15:20 **EvoBatch: A Mini-Batch, Compute-Constrained Evolutionary Algorithm Reduces the Generalization Gap in Deep Learning** 
Toki Inan, Shahana Shultana, Amarda Shehu 
- 15:20–15:40 **★ Mortar: Evolving Mechanics for Automatic Game Design** 
Muhammad Nasir, Yuchen Li, Steven James, Julian Togelius 
- 15:40–16:00 **★ Distributional Value Estimation Without Target Networks for Robust Quality-Diversity** 
Behrad Koochy, Jamie Bayne 
- 16:00–16:20 **COvolve: Adversarial Co-Evolution of Large-Language-Model-Generated Policies and Environments via Two-Player Zero-Sum Game** 
Alkis Sygkounas, Rishi Hazra, Andreas Persson, Pedro Zuidberg dos Martires, Amy Loutfi 

ECOM BP

Manuel Antonio

Chair: Enrique Alba, *University of Málaga, Spain*

- 15:00–15:20 **★ Obtaining Partition Crossover masks using Statistical Linkage Learning for solving noised optimization problems with hidden variable dependency structure** 
Michał Przewozniczek, Bartosz Frej, Marcin Komarnicki, Michał Prusik, Renato Tinós 
- 15:20–15:40 **★ Vehicle Routing Problem with Resource-Constrained Pickup and Delivery: A Heuristic-Informed BRKGA with Pattern-Based Analysis** 
Manbir Sodhi, Romesh Prasad 
- 15:40–16:00 **★ A Heterogeneous Ant Colony Framework with Semantic Manifold Learning for Dynamic Ore-Flow Blending** 
Hao Tang, Changhe Li, Guoyu Chen, Shoufei Han, Michalis Mavrovouniotis, Miqing Li 

BBSR + ENUM BP

Poas

Chair: Anne Auger, *Inria; CMAP, Ecole Polytechnique, France*

- 15:00–15:20 **Accelerating Black-Box Bilevel Optimization with Rank-Based Upper-Level Value Function Approximation** 
Marc Ong, Youhei Akimoto 
- 15:20–15:40 **★ S3-LRA-xNES: Step-size and Shape Separated Learning Rate Adaptation for Exponential Natural Evolution Strategy** 
Kosuke Ujihara, Masahiro Nomura, Isao Ono 
- 15:40–16:00 **★ Assessing Reproducibility in Evolutionary Computation: A Case Study using Human- and LLM-based Assessment** 
Francesca Da Ros, Tarik Začiragić, Aske Plaat, Thomas Bäck, Niki van Stein 
- 16:00–16:20 **★ Exploration of Pareto-preserving search-space transformations in multi-objective test functions** 
Diederick Vermetten, Jeroen Rook 

EMO 2

Monteverde

Chair: Carlos Ignacio Hernández Castellanos, *IIMAS-UNAM, Mexico*

- 15:00–15:20 **On Reference Set Selection for Constrained Multiobjective Optimization Problems** 
Luka Opravš, Dimo Brockhoff, Tea Tušar 
- 15:20–15:40 **Weight Vector Specification in MOEA/D to Find Balanced and Promising Solutions for Multi-Criteria Decision Making** 
Lie Meng Pang, Michael Emmerich, Hisao Ishibuchi 

- 15:40–16:00 **On the Use of Bi-Objective Evolutionary Algorithms for the Stochastic Multiple Knapsack Problem under Dynamic Constraints** 
Ishara Hewa Pathirana, Aneta Neumann  
- 16:00–16:20 **Adaptive Ensemble Multi-Objective Differential Evolution for High-Dimensional Gene Selection** 
Dhia eddine Bouazizi, Asma Amdouni, Daniel Rodriguez, ghaith Manita  

GECH 1

Tortuguero

Chair: Yuri Lavinas, *IRIT - CNRS UMR5505, University of Toulouse, France*

- 15:00–15:20 **An Efficient Hybrid Racing Method for Portfolio Configuration** 
Anthony Rasulo, Manuel López-Ibáñez, Julia Handl, Mario Andrés Muñoz, Kate Smith-Miles  
- 15:20–15:40 **Automatically Designing Novel Evolutionary Architectures** 
Braden Tisdale, Daniel R. Tauritz, Aaron Pope  
- 15:40–16:00 **Digital Red Queen: Adversarial Program Evolution in Core War with LLMs** 
Akarsh Kumar, Ryan Bahlous-Boldi, Prafull Sharma, Phillip Isola, Sebastian Risi, Yujin Tang, David Ha  
- 16:00–16:20 **Discrete Self-Adaptation in Competitive Coevolution for Constrained Hardware** 
Steven Jorgensen, Jennifer Fairhurst, Erik Hemberg, Una-May O'Reilly  

RWA 3

Chirripó

Chair: Sylvain Cussat-Blanc, *University of Toulouse, IRIT - CRNS UMR5505 - ANITI, France*

- 15:00–15:20 **Integrating Digital Twins into Evolutionary Optimization of Factory Layouts** 
Jiri Kubalik, Antonin Miller, Petr Kadera  
- 15:20–15:40 **Genetic Programming for Evolving Tailored Spectral Vegetation Indices in the Context of Crop Pathogen Detection** 
Anthony Stein, Johanna Vaske, Simon Mielke  
- 15:40–16:00 **Quantum-Behaved PSO-DE for Ubiquitous Connectivity in Space-Terrestrial Cell-Free Massive MIMO Systems** 
Anh-Thu Ngo Tran, Chien Trinh  
- 16:00–16:20 **Survival Is Not Enough: Improving Glycemic Control with Sexual Selection** 
Nuno Lourenço, Óscar Garnica, J. Ignacio Hidalgo  

GP 2 + Impact

Gandoca-Manzanillo

Chair: Nelishia Pillay, *University of Pretoria, South Africa*

- 15:00–15:20 ☆ **The role of genetic programming in agentic AI – reflecting on 10 years of TPOT**
Jason Moore, Randal Olson, Ryan Urbanowicz  
- 15:20–15:40 **Teaching the Teacher: The Role of Teacher-Student Smoothness Alignment in Genetic Programming-based Symbolic Distillation** 
Soumyadeep Dhar, Kei Sen Fong, Mehul Motani  
- 15:40–16:00 **A Boundary-Aware Multi-Objective Genetic Programming Classifier with Stacking-Based Integration** 
Kaiwei Yuan, Xiaoying Gao, Jianbin Ma, Jia Zhao  
- 16:00–16:20 **Evolving Dispatching Rules for the Unrelated Parallel Machines Scheduling Problem with Precedence and Resource Constraints with Genetic Programming** 
Josipa Sabljo, Mateja Dumić, Marko Durasević  

Thursday, July 16 | 17:00–18:00

L4EC BP

Corcovado

Chair: Lee Spector, *Amherst College; UMass Amherst, Hampshire College, United States*

- 17:00–17:20 **LLaMEA-BO: A Large Language Model Evolutionary Algorithm for Automatically Generating Bayesian Optimization Algorithms** 
Wenhu Li, Niki van Stein, Thomas Bäck, Elena Raponi  
- 17:20–17:40 **★ Deep Learning for Feature-Free Landscape Analysis in Combinatorial Optimization: An Empirical Study on NK-landscapes** 
Sarah Degaugue, Bilel Derbel  
- 17:40–18:00 **★ A Learning-Based Cooperative Coevolution Framework for Heterogeneous Large-Scale Global Optimization** 
Wenjie Qiu, Zixin Wang, Hongyu Fang, Zeyuan Ma, Yue-Jiao Gong  

BBSR 2

Poas

Chair: Tea Tusar, *Jožef Stefan Institute, Jožef Stefan International Postgraduate School, Slovenia*

- 17:00–17:20 **Benchmarking Interactive Multi-Criterion Decision-Making Procedures Following Machine Learning-Based Decision-Maker** 
Deepanshu Yadav, Kalyanmoy Deb  
- 17:20–17:40 **Constructing Streams of Optimization Instances for Benchmarking Algorithm Selection and Configuration Approaches in Streaming Scenarios** 
Margherita Battistotti, Manuel López-Ibáñez, Kate Smith-Miles, Julia Handl, Mario Muñoz  
- 17:40–18:00 **Designing Landscape-Aware Benchmarks with Explicit Local Optima for Single- and Multi-Objective Optimization** 
Shuhei Tanaka, Shoichiro Tanaka, Kippei Mizuta, Toshiharu Hatanaka  

CS 1

Arenal

Chair: Pier Luca Lanzi, *Politecnico di Milano, Italy*

- 17:00–17:20 **Social Learning Strategies for Evolved Virtual Soft Robots** 
Ege de Bruin, Kyrre Glette, Kai Ellefsen, Giorgia Nadizar, Eric Medvet  
- 17:20–17:40 **Tournament Informed Adversarial Quality Diversity** 
Timothée Anne, Noah Syrkis, Meriem Elhosni, Florian Turati, Alexandre Manai, Franck Legendre, Alain Jaquier, Sebastian Risi  
- 17:40–18:00 **Emergence Of Internal State-Modulated Swarming In Multi-Agent Patch Foraging System** 
Siddharth Chaturvedi, Ahmed El-Gazzar, Marcel Gerven  

EMO 3

Monteverde

Chair: Saúl Zapotecas-Martínez, *Instituto Nacional de Astrofísica, Óptica y Electrónica, Mexico*

- 17:00–17:20 **Deep Reinforcement Learning Based Environmental and Mating Selection for Evolutionary Multi-objective Optimization** 
Cheng-Je Tsai, Tsung-Che Chiang  
- 17:20–17:40 **P-MOEB: A Pairwise Evolutionary Framework for Balanced Clustering** 
Jianhao Zhu, Yunhui Liang, Yan Chen, Peng Wu, Chao Wen, Tao Yan, Peng Zhou, liang Du  

SI 1

Tortuguero

Chair: Pierrick Legrand, *Bordeaux INP, ENSC, F-33400 Talence, France; Inria Center of the University of Bordeaux, France, France*

- 17:00–17:20 **A Hybrid GA-PSO with Collective Learning for Obstacle-Aware Sensor Node Deployment in IoT-Enabled Networks** 
Nguyen Binh, Ho Duc Luong, Nguyen Tuan, Le Duan, Huynh Binh  

- 17:20–17:40 **Particle Swarm Optimization with Population Dynamics – Artificial Splitting, Extinction, and Migration** 
Yutong Zou, Wenjun Wang, Wei Pang  
- 17:40–18:00 **Efficient EV Charging Allocation in Fog Computing via Committee-Based Surrogate-Assisted PSO** 
Ibtissem Mokni, Sonia Yassa, Stéphane Zuckerman, Olivier Romain, Mohamed nazih Omri
 

RWA 4

Chirripó

Chair: Elizabeth Wanner, *Aston University, United Kingdom*

- 17:00–17:20 **Dual-Layer Synchronous Ring Cellular UMDA Based On Anisotropic RBF Surrogate for High-Dimensional Evolutionary Optimization in Energy Management** 
Jingao Liu, Chixin Xiao, Dechen Jiang, Maoxin He, Limei Liu  
- 17:20–17:40 **Anisotropic Random-Feature Surrogate-Assisted Evolutionary Algorithm for High-Dimensional Risk-Based Energy Management Optimization** 
Jiaming Hu, Chixin Xiao, Limei Liu, Maoxin He, Dechen Jiang  
- 17:40–18:00 **Robust Evolutionary Optimization for Demand-Balanced Railway Route Design** 
George Brooks, Sean Walton, Markus Roggenbach, John Armstrong, Alma Rahat  

NE 1

Gandoca-Manzanillo

Chair: Travis Desell, *Rochester Institute of Technology, United States*

- 17:00–17:20 **Measuring Intrinsic Dimension of Multiobjective Landscapes and Dimensionality-Reduced Neuroevolution of Deep Reinforcement Learning** 
oladayo Ajani, Jeonggeun Kim, Jong Lee, Byungchul Tak  
- 17:20–17:40 **Asynchronous Neuro-Evolutionary Symbolic Regression with Maturity-Based Replacement** 
Jiri Kubalík, Naďa Fučelová, Robert Babuška  
- 17:40–18:00 **Investigating Memetic Scheduling Strategies for Distributed Neuroevolution** 
Diana Velychko, Travis Desell  

Friday, July 17 | 09:00–10:30**RWA 5 + EML 4**

Corcovado

Chair: Vanessa Volz, *Centrum Wiskunde & Informatica (CWI), Netherlands*

- 09:00–09:20 **When Fitness Is Cheap: Pareto-Based Evolutionary Optimisation for Lightweight Neural Architectures** 
Jingyue Cong, Kai Wei, Yameng Peng, Yanan Sun, Andy Song  
- 09:20–09:40 **Search-Based Risk Feature Discovery in Document Structure Spaces under a Constrained Budget** 
Saisubramaniam Gopalakrishnan, Harikrishnan P M, Dagnachew Birru  
- 09:40–10:00 **CURE: Adaptive Multi-Channel Fusion via Genetic Optimization for Chronic Wound Image Analysis**
Agata Maria Wijata, Maria Bienkowska  
- 10:00–10:20 **Evolutionary-Driven Bayesian Optimization for Automated Molecular Docking with AutoDock Vina** 
Alejandro Lopez-Rincon, Brigitta Varga, David Rojas-Velazquez, Alberto Tonda  

NE 2 + GP 3

Manuel Antonio

Chair: Thomas Helmuth, *Hamilton College, United States*

- 09:00–09:20 **EvoTS: Evolutionary Transformer Search for Time Series Forecasting** 
AbdElRahman ElSaid, Damir Pulatov  

- 09:20–09:40 **LLM-ODE: Data-driven Discovery of Dynamical Systems with Large Language Models** 
Amirmohammad Ziaei Bideh, Jonathan Gryak  
- 09:40–10:00 **Diversified Residual Symbolic Regression** 
Koki Ikeda, Masahiro Nomura, Ryoki Hamano  
- 10:00–10:30 **ROIDS: Robust Outlier-Aware Informed Down-Sampling** 
Alina Geiger, Martin Briesch, Dominik Sobania, Franz Rothlauf  

GECH 2

Poas

Chair: Ilya Basin, *Noblis, United States*

- 09:00–09:20 **ProFiT: Program Search for Financial Trading** 
Matthew Siper, Ahmed Khalifa, Lisa Soros, Muhammad Nasir, Juyan Azhang, Julian Togelius  
- 09:20–09:40 **Energy-Aware Metaheuristics** 
Enrique Alba, Tomohiro Harada, Gabriel Luque  
- 09:40–10:00 **Interactive LLM-assisted curriculum learning for multi-task policy search** 
Berfin Sakallioğlu, Giorgia Nadizar, Eric Medvet  
- 10:00–10:20 **MixDiBB: Distributed Black-Box Optimization for Mixed-Parameter Search Spaces** 
Luca Rolshoven, Matthias Stürmer, Giuseppe Cuccu  

EMO 4

Monteverde

Chair: Manuel López-Ibáñez, *University of Manchester, United Kingdom*

- 09:00–09:20 **An Adaptive Offspring Generation Wrapper for Constrained Multimodal Multiobjective Evolutionary Algorithms** 
Jiali Gu, Yu Sun  
- 09:20–09:40 **Pareto Optimization with Robust Evaluation for Noisy Subset Selection** 
Yiheng Xu, Danxuan Liu, Bin Zhang, Weiyong Yang, Chao Qian  
- 09:40–10:00 **Approximating the Hypervolume Indicator using Fast Low-Discrepancy Sequences** 
Manuel López-Ibáñez  
- 10:00–10:20 **Bi-objective Stochastic Simulation Optimization on Integer Lattices via Scalarization** 
Sebastian Rojas Gonzalez, Ivo Couckuyt, Joshua Knowles  

ECOM 3 + GA 2

Tortuguero

Chair: Niki van Stein, *Leiden University, Netherlands*

- 09:00–09:20 **LLM Policy Induction for Heuristic Search Control: A Trace-Driven ALNS Framework** 
Tam Nguyen  
- 09:20–09:40 **Interpretable Algorithm Selection via Multi-Objective Evolutionary Decision Trees** 
Matheus Vilas Boas, Elizabeth Wanner, Gladston Juliano Moreira  
- 09:40–10:00 **The hop-like problem nature – unveiling and modelling new features of real-world problems** 
Michał Przewozniczek, Bartosz Frej, Marcin Komarnicki  
- 10:00–10:20 **Evolutionary Tunneling and Periodicity Across the Big Valley Distribution** 
Darrell Whitley, Gabriela Ochoa, Francisco Chicano  

RWA 6

Chirripó

Chair: Yazmin Maldonado, *Instituto Tecnológico de Tijuana, Mexico*

- 09:00–09:20 **Evolutionary Multitask Optimization for Coupled Continuous Decision Spaces: A GPU-Accelerated Pre-optimization for Movable-Antenna-and-RIS-aided Systems** 
Ho Duc Luong, Tran Dung, Lang Nguyet Anh, Trinh Chien  
- 09:20–09:40 **Adaptive Grid-Based Multi-Objective Evolutionary Optimization for Coordinated Truck-Drone Routing with Backhauls** 
Bui Xuan Son, Nguyen Thi Ha, Nguyen Tien Dang, Bui Trong Duc, Huynh Thi Thanh Binh
 
- 09:40–10:00 **Evolutionary Computation for Tax-Minimizing Strategies in Special Economic Zones** 
Andres Leguizamon, Carlos Sanchez, Sofia Ocampo, Una-May O'Reilly, Erik Hemberg  
- 10:00–10:20 **Hybridized Evolution Accelerating Thermostability (HEAT): A Novel MOEA Approach to EASME Protein Engineering** 
James Browning, Daniel R. Tauritz, John Beckmann  

L4EC 2

Gandoca-Manzanillo

Chair: Elena Raponi, *Leiden University, Netherlands*

- 09:00–09:20 **Advancing CMA-ES with Learning-Based Cooperative Coevolution for Scalable Optimization** 
Wenjie Qiu, Hao Guo, Zeyuan Ma, Yue-Jiao Gong  
- 09:20–09:40 **Detect and Act: Automated Dynamic Optimizer through Meta-Black-Box Optimization** 
Zijian Gao, Zeyuan Ma, Yuanting Zhong, Yue-Jiao Gong, Hongshu Guo  
- 09:40–10:00 **Does Dimensionality Reduction via Random Projections preserve Landscape Features?** 
Ivan Olarte Rodriguez, Anja Janković, Thomas Bäck, Elena Raponi  